THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 1: **A Square and a Cube** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The smallest number by which 72 must be multiplied to get a perfect square is: 1

(a) 2 **(b)** 1 **(c)** 4 **(d)** 3

2. The smallest number by which 192 must be divided to get a perfect cube is: 1

(a) 6 **(b)** 4 **(c)** 3 **(d)** 5

3. The ones digit of 127^2 is: 1

(a) 9 **(b)** 7 **(c)** 1 **(d)** 4

4. $\sqrt{1.44}$ is equal to: 1

(a) 12.0 **(b)** 1.3 **(c)** 1.2 **(d)** 0.12

5. $\sqrt[3]{0.027}$ is equal to: 1

(a) 0.3 **(b)** 0.4 **(c)** 3.0 **(d)** 0.03

6. The value of 45^2 is: 1

(a) 2000 **(b)** 2030 **(c)** 2050 **(d)** 2025

7. The smallest number by which 98 must be multiplied to get a perfect square is: 1

(a) 3 **(b)** 1 **(c)** 4 **(d)** 2

8. The smallest number by which 250 must be divided to get a perfect cube is: 1

(a) 1 **(b)** 4 **(c)** 2 **(d)** 3

9. The ones digit of 238^2 is: 1

(a) 6 **(b)** 4 **(c)** 8 **(d)** 2

10. $\sqrt{2.25}$ is equal to: 1

(a) 1.6 **(b)** 15.0 **(c)** 1.5 **(d)** 0.15

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find $\sqrt{2916}$ by prime factorisation. 2

12. Evaluate $\sqrt[3]{\frac{27}{512}}$. 2

13. Evaluate $\sqrt{\frac{81}{256}}$. 2

14. Find $\sqrt{4624}$ by prime factorisation. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the smallest number by which 1872 must be multiplied to make it a perfect square. Find the square root of the product. 3

16. Find the smallest number by which 12000 must be divided to make it a perfect cube. Find the cube root of the quotient. 3

17. Find $\sqrt{1.96}$ using prime factorisation of the whole number obtained after removing the decimal point. 3

OR

Find the smallest number by which 2646 must be multiplied to make it a perfect square. Find the square root of the product. 3

18. Find the smallest number by which 10976 must be divided to make it a perfect cube. Find the cube root of the quotient. 3

OR

Find $\sqrt{3.24}$ using prime factorisation of the whole number obtained after removing the decimal point. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A school has 1500 students who are to stand in rows so that the number of rows equals the number of students in each row. How many more students are needed to complete the arrangement? What is the number of rows then? 5

OR

The volume of a cube is 1728 cm³. Find the side of the cube and its total surface area. 5

20. A school has 2000 students who are to stand in rows so that the number of rows equals the number of students in each row. How many more students are needed to complete the arrangement? What is the number of rows then? 5

OR

The volume of a cube is 2744 cm³. Find the side of the cube and its total surface area. 5

END OF QUESTION PAPER — Total: 40 Marks

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